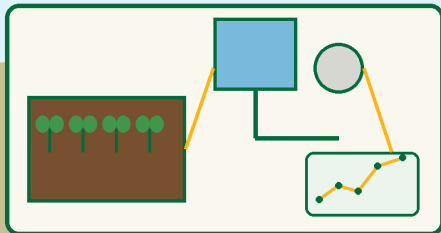


SYSTEM ARCHITECTURE

ARCHITECTURE



ARCHITECTURE

SYSTEM ARCHITECTURE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

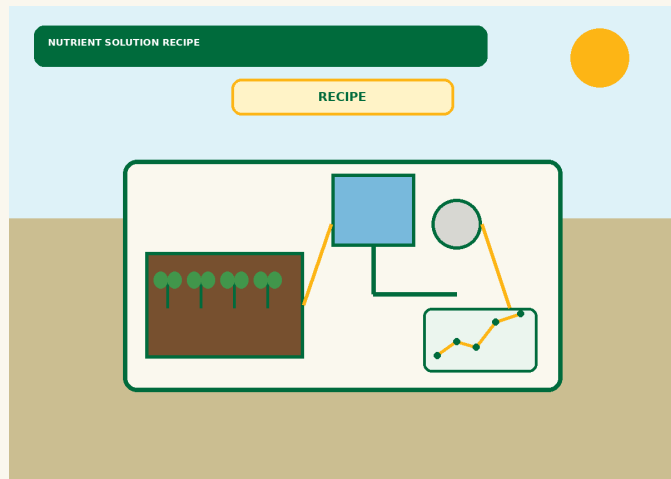
TECHNICAL CONCEPT

Soilless system architecture defines reservoirs, pumps, delivery lines, root zones, drainage, aeration, sensors, and return pathways.

ANALYSIS / DESIGN TASK

Create a functional system diagram.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



RECIPE

NUTRIENT SOLUTION RECIPE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

CARD 902 • NUTRIENT SOLUTION RECIPE

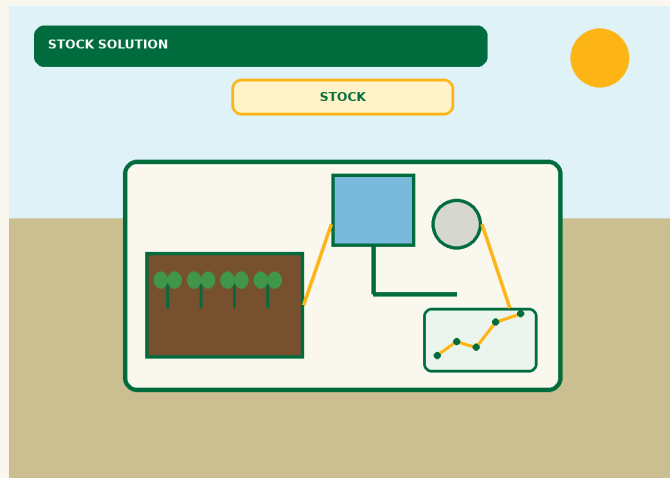
TECHNICAL CONCEPT

A nutrient-solution recipe specifies target concentrations of essential mineral nutrients and must account for source water and crop requirements.

ANALYSIS / DESIGN TASK

Interpret a teacher-provided formulation.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



STOCK

STOCK SOLUTION

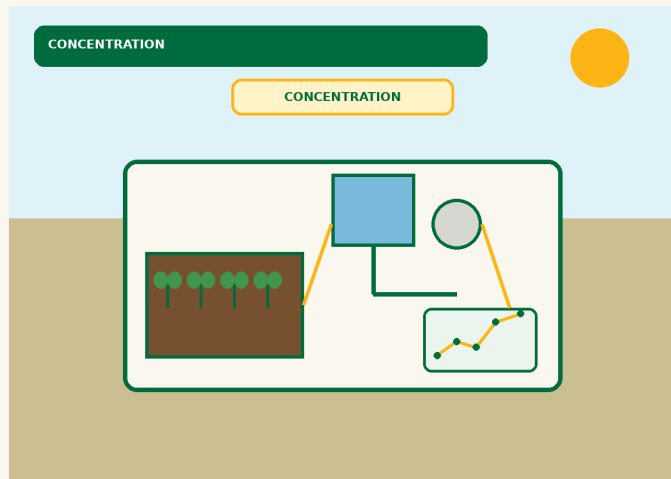
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A stock solution is a concentrated nutrient mixture diluted to produce the final working solution.

ANALYSIS / DESIGN TASK

Calculate a safe dilution from provided values.



CONCENTRATION

CONCENTRATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

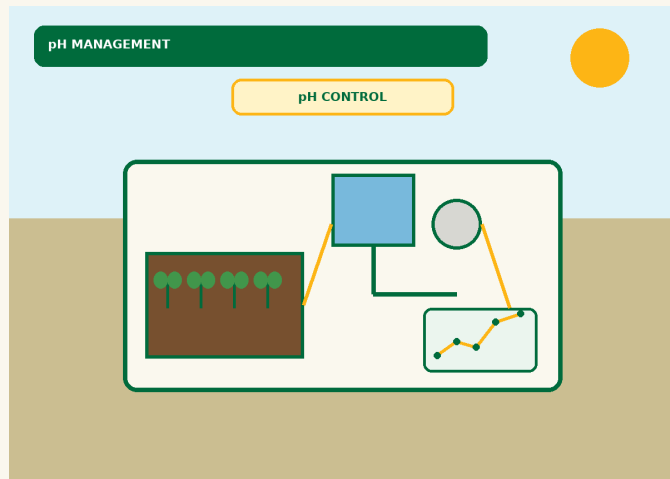
TECHNICAL CONCEPT

Solution concentration can be expressed in mass per volume, molar units, or other defined bases depending on the nutrient-management task.

ANALYSIS / DESIGN TASK

Convert a simple mg/L concentration.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



pH CONTROL

pH MANAGEMENT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

pH management maintains solution acidity or alkalinity within an appropriate operating range for nutrient availability and crop performance.

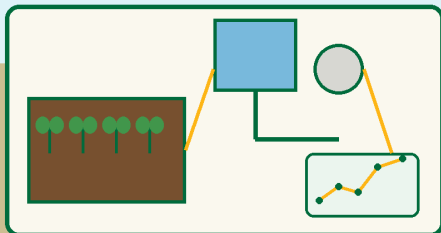
ANALYSIS / DESIGN TASK

Interpret a pH trend and identify when action is indicated.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

ELECTRICAL CONDUCTIVITY

EC



EC

ELECTRICAL CONDUCTIVITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

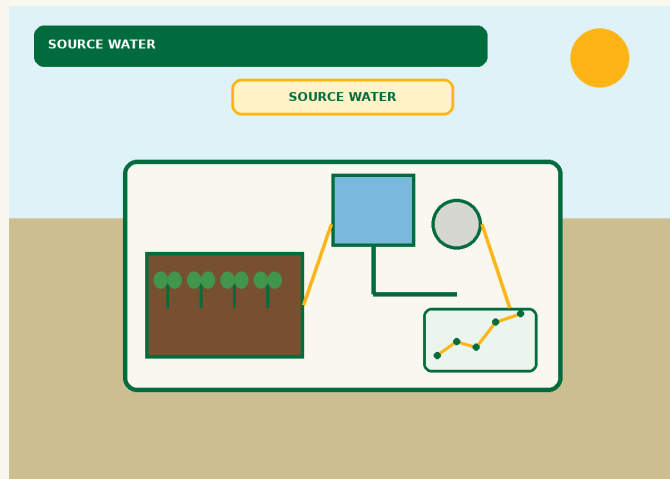
TECHNICAL CONCEPT

Electrical conductivity is an operational indicator of dissolved ionic concentration but does not identify individual nutrients.

ANALYSIS / DESIGN TASK

Interpret EC changes over time.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



SOURCE WATER

SOURCE WATER

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

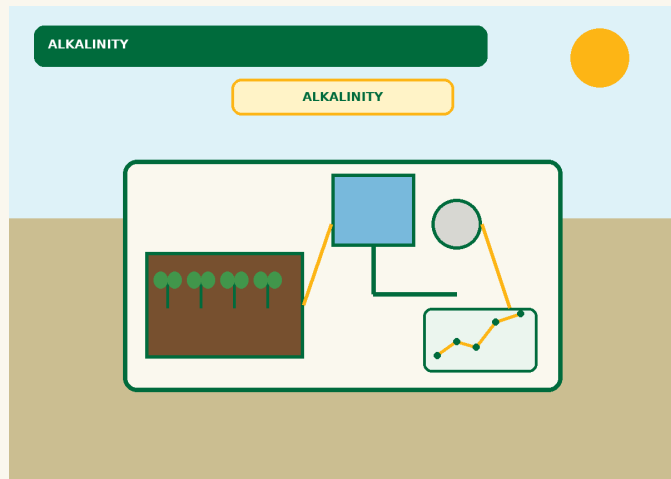
TECHNICAL CONCEPT

Source-water chemistry contributes dissolved ions and alkalinity that influence nutrient formulation and pH management.

ANALYSIS / DESIGN TASK

Review a simplified source-water report.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



ALKALINITY

ALKALINITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Alkalinity describes water's acid-neutralizing capacity and strongly influences how pH responds to acid addition or plant uptake.

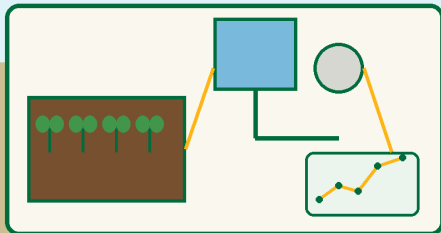
ANALYSIS / DESIGN TASK

Interpret provided alkalinity categories.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

DISSOLVED OXYGEN

DO



DO

DISSOLVED OXYGEN

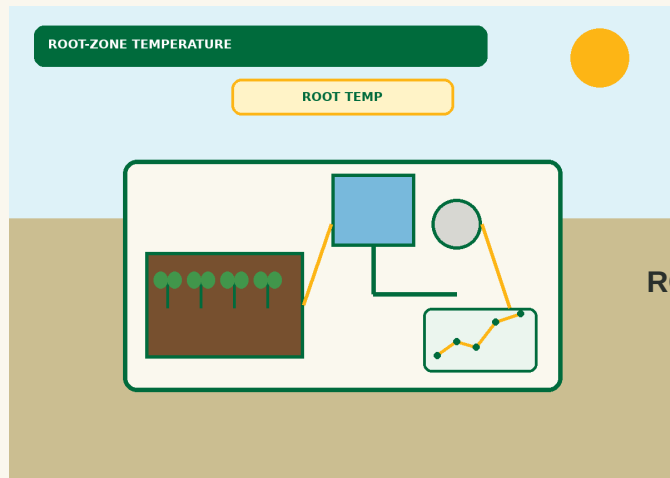
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Dissolved oxygen is oxygen present in solution and is required for healthy root respiration in water-saturated systems.

ANALYSIS / DESIGN TASK

Interpret a DO reading.



ROOT TEMP

ROOT-ZONE TEMPERATURE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

CARD 910 • ROOT-ZONE TEMPERATURE

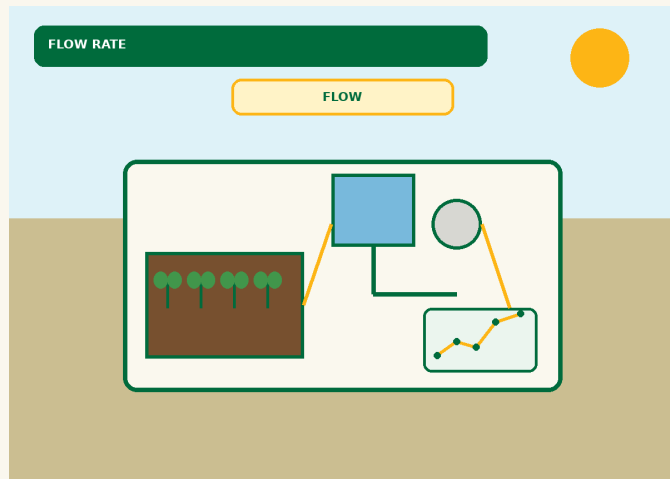
TECHNICAL CONCEPT

Root-zone temperature affects respiration, oxygen solubility, nutrient uptake, and root biological processes.

ANALYSIS / DESIGN TASK

Analyze a daily temperature series.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



FLOW

FLOW RATE

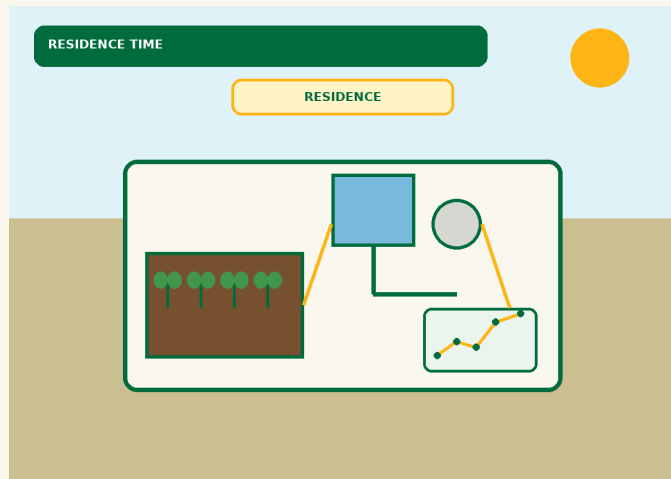
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Flow rate influences nutrient transport, oxygen delivery, residence time, and hydraulic performance in circulating systems.

ANALYSIS / DESIGN TASK

Measure and compare channel flow.



RESIDENCE

RESIDENCE TIME

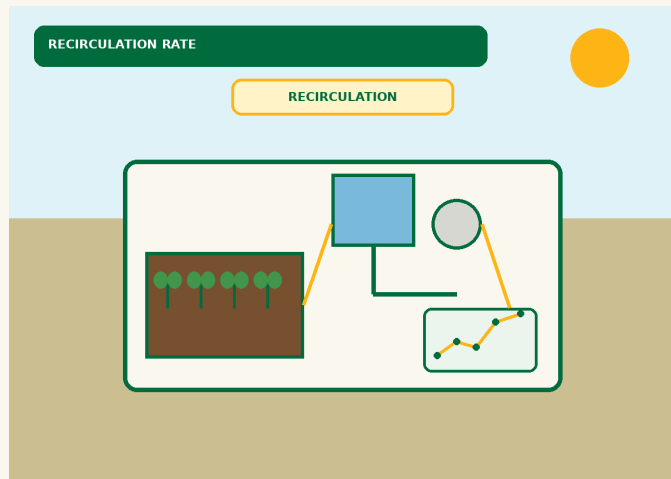
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Hydraulic residence time describes how long solution remains within a defined tank or system volume at a given flow rate.

ANALYSIS / DESIGN TASK

Calculate a simplified residence time.



RECIRCULATION

RECIRCULATION RATE

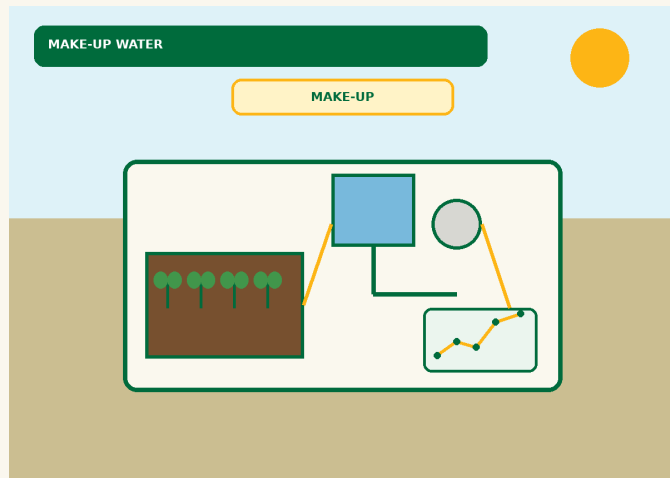
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Recirculation rate describes how frequently system solution is moved through the root-zone circuit.

ANALYSIS / DESIGN TASK

Calculate reservoir turnovers per hour.



MAKE-UP

MAKE-UP WATER

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

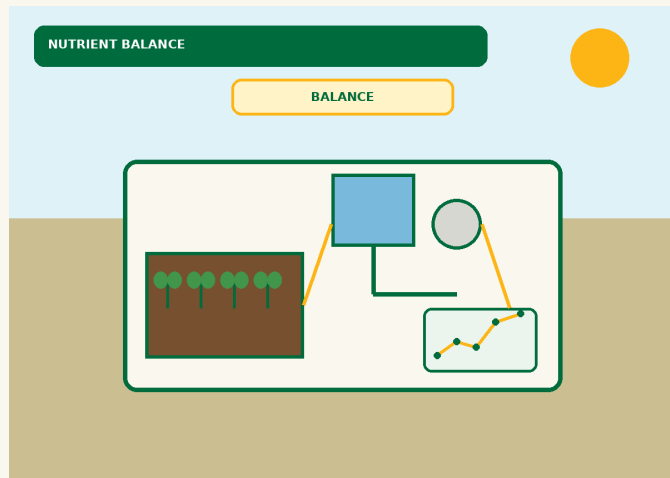
TECHNICAL CONCEPT

Make-up water replaces solution volume lost through crop uptake, evaporation, sampling, leakage, or maintenance.

ANALYSIS / DESIGN TASK

Quantify daily make-up volume.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



BALANCE

NUTRIENT BALANCE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

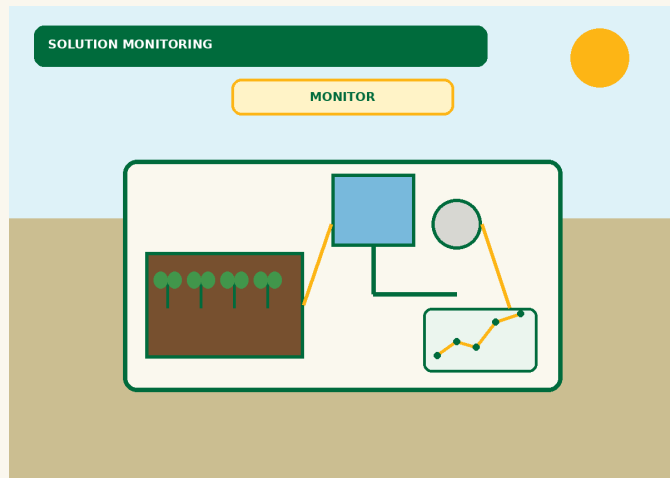
TECHNICAL CONCEPT

A nutrient balance tracks nutrient inputs, plant uptake, solution storage, discharge, and other losses within a defined system.

ANALYSIS / DESIGN TASK

Complete a simplified nutrient balance.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



MONITOR

SOLUTION MONITORING

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

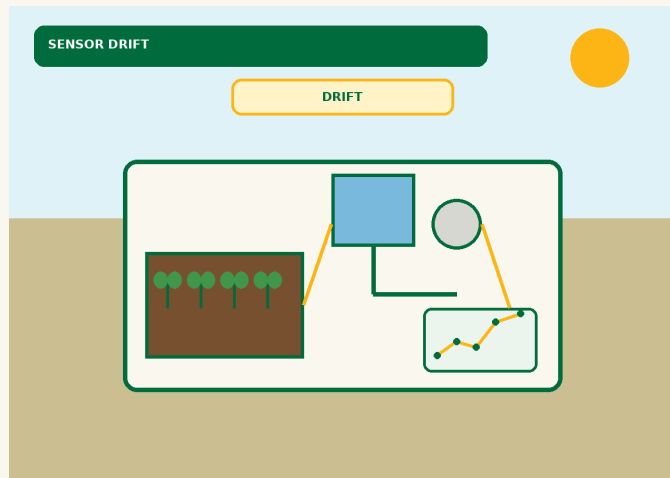
TECHNICAL CONCEPT

Solution monitoring integrates pH, EC, temperature, dissolved oxygen, level, and flow to assess operating condition.

ANALYSIS / DESIGN TASK

Interpret a monitoring dashboard.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



DRIFT

SENSOR DRIFT

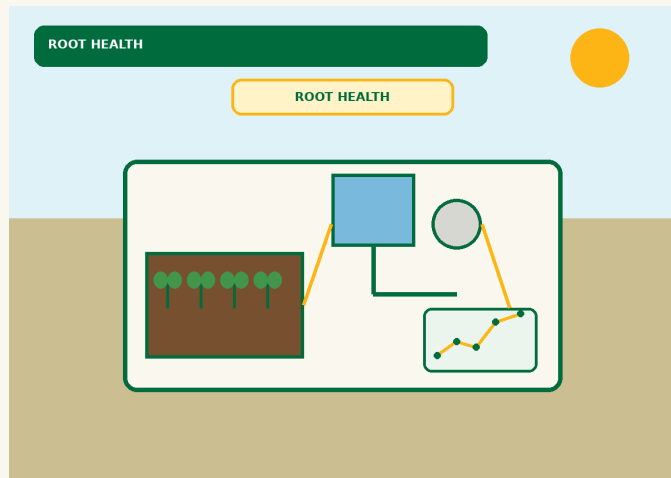
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Sensor drift is a gradual change in instrument response that can create biased readings without calibration or validation checks.

ANALYSIS / DESIGN TASK

Detect possible drift from reference checks.



ROOT HEALTH

ROOT HEALTH

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Root-health assessment considers color, branching, odor, texture, disease symptoms, oxygen, temperature, and solution conditions.

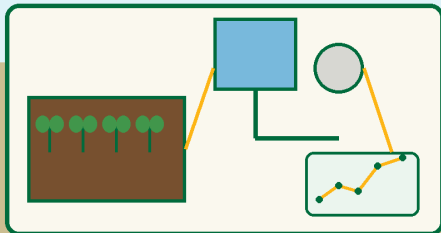
ANALYSIS / DESIGN TASK

Evaluate a root-zone case study.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

TROUBLESHOOTING MATRIX

DIAGNOSE



DIAGNOSE

TROUBLESHOOTING MATRIX

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A troubleshooting matrix links symptoms and measurements to plausible causes and verification tests before corrective action.

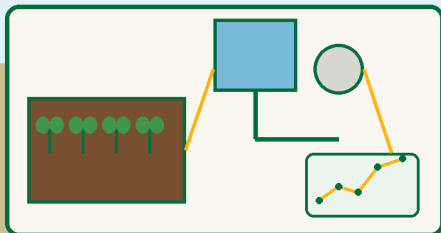
ANALYSIS / DESIGN TASK

Build a cause-test-action table.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

SOILLESS OPTIMIZATION

OPTIMIZE



OPTIMIZE

SOILLESS OPTIMIZATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Soilless optimization balances crop performance, nutrient solution, water, energy, monitoring, reliability, and operating cost.

ANALYSIS / DESIGN TASK

Recommend one controlled optimization experiment.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE